

[illegible]

Component Tray Layout

USB PCIe Expansion Card · Rev 0.1

Tray: 150×150×10mm · 11×11 compartments (10 dividers/axis, 1mm thick)

Generated 2026-09-10

LOOKUP TABLE

Cell number on the tray → what's in it. Grouped in the same reading order the tray is filled in: left→right, top→bottom, one row at a time.

#	CATEGORY	REFERENCE(S)	VALUE / PART	PACKAGE	QTY
1	RESISTOR	R1, R12–R15	0Ω jumper	0603	5
2	RESISTOR	R2,3,5,6,7,16,23,27–30,40	10kΩ	0603	12
3	RESISTOR	R4, R24	100kΩ	0603	2
4	RESISTOR	R8	1.6kΩ	0402	1
5	RESISTOR	R9	680Ω	0402	1
6	RESISTOR	R10, R11, R17, R26, R31	0Ω jumper	1206	5
7	RESISTOR	R18	47kΩ	0603	1
8	RESISTOR	R19	11kΩ	0402	1
9	RESISTOR	R20	17.4kΩ	0402	1
10	RESISTOR	R21	34.8kΩ	0402	1
11	RESISTOR	R22	1MΩ	0603	1
12	RESISTOR	R25	133Ω	0603	1
13	RESISTOR	R32–R39	1kΩ	0603	8
14	CERAMIC CAPACITOR	C1–C17	0.01μF	0402	17
15	CERAMIC CAPACITOR	C18–27,29,37–44	0.1μF	0402	19
16	CERAMIC CAPACITOR	C28,62,63,66,69,74,75	1μF	0603	7

#	CATEGORY	REFERENCE(S)	VALUE / PART	PACKAGE	QTY
17	CERAMIC CAPACITOR	C30, C31	8pF	0402	2
18	CERAMIC CAPACITOR	C32,34,64,67,70,71	10μF	0805	6
19	CERAMIC CAPACITOR	C33,36,50,57,59	0.1μF	0603	5
20	CERAMIC CAPACITOR	C35, C51, C56	10μF	1206	3
21	CERAMIC CAPACITOR	C45, C46	6.8μF	1206	2
22	CERAMIC CAPACITOR	C49	10nF	1206	1
23	CERAMIC CAPACITOR	C52	1.8nF	0402	1
24	ELECTROLYTIC CAPACITOR	C47	22μF electrolytic	D6.3×L6.1	1
25	CERAMIC CAPACITOR	C53	47pF	0402	1
26	CERAMIC CAPACITOR	C58, C60, C61	22μF	0603	3
27	DIODE / LED	D1	1SS294 Schottky	SC-59	1
28	ELECTROLYTIC CAPACITOR	C48	180μF electrolytic	D8×L10.5	1
29	DIODE / LED	D2	B560C Schottky	DO-214AB	1
30	DIODE / LED	D3	RSX501LAM20TR	SOD-128	1
31	ELECTROLYTIC CAPACITOR	C65,68,72,73	150μF Al-polymer	D8×L6.9	4
32	DIODE / LED	D4, D5	RBR2MM30BTR	SOD-123F	2
33	CERAMIC CAPACITOR	C54, C55	100μF	1210	2
34	DIODE / LED	D6	B320A Schottky	DO-214AC	1
35	INDUCTOR / FERRITE	L2	1μH inductor	SPM6530	1

#	CATEGORY	REFERENCE(S)	VALUE / PART	PACKAGE	QTY
36	DIODE / LED	D7–D10	White LED	0805	4
37	DIODE / LED	D11–D13	Blue LED	0805	3
38	DIODE / LED	D14, D15	Yellow LED	0805	2
39	INDUCTOR / FERRITE	L1	3.9µH inductor	SRU8043	1
40	RESISTOR	F1	3.15A fuse	0603	1
41	INDUCTOR / FERRITE	FB1–FB5	Ferrite bead	0805	5
42	INDUCTOR / FERRITE	L3	1.5µH inductor	1008	1
43	TRANSISTOR / MOSFET	Q1	2SA1576 PNP	SOT-323	1
44	TRANSISTOR / MOSFET	Q2	2SC4081 NPN	SOT-323	1
45	IC / CONNECTOR / CRYSTAL	U6	641737-1 header	THT 4-pos	1
46	TRANSISTOR / MOSFET	Q4	DMT6016LSS N-MOS	SO-8	1
47	TRANSISTOR / MOSFET	Q5	SI4134DY N-MOS	SO-8	1
48	IC / CONNECTOR / CRYSTAL	U5	RT5762 buck reg	SOT-563	1
49	IC / CONNECTOR / CRYSTAL	U4	LTC3872 boost ctrl	TSOT-23-8	1
50	IC / CONNECTOR / CRYSTAL	U3	LM22678 buck reg	TO-263-7	1
51	IC / CONNECTOR / CRYSTAL	U7	AP7215 LDO	SOT-89-3	1
52	IC / CONNECTOR / CRYSTAL	U2	MX25L4006E flash	SOP-8	1
53	IC / CONNECTOR / CRYSTAL	U1	UPD720201 host ctrl ⚠ ESD	QFN68 8×8	1

#	CATEGORY	REFERENCE(S)	VALUE / PART	PACKAGE	QTY
54	IC / CONNECTOR / CRYSTAL	U12–U15	SP3011 TVS array	UDFN14	4
55	IC / CONNECTOR / CRYSTAL	U8–U11	AP2191 power switch	SOP-8	4
56	IC / CONNECTOR / CRYSTAL	X1	24MHz crystal	4-SMD 3.2×2.5	1

NOT IN THE TRAY

- **J2–J5** (USB3-A stacked connectors) — taller than the 10mm box even lying flat, and there are only four. Keep them in their original packaging.
- **TP1–TP7** (test points) — generic through-hole pins, not sorted by value, no tray cell needed.

HANDLE WITH CARE — CELL 53

U1 (Renesas UPD720201, QFN68) is the single most expensive and ESD-sensitive part in this BOM. Its cell is outlined in copper on the printed label — keep it away from the edge of your bench and handle by the packaging tape, not bare-handed, until it's ready to place.

ONE ASSUMPTION WORTH CHECKING

Only the X-divider positions were given, so the Y-axis is assumed to mirror them: 10 dividers, 1mm thick, across a 148mm interior → 11 equal rows of ~12.5mm. If your Y-axis differs, the row heights on the cut-out above will be off and it's worth re-exporting this label.